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Freshwater marshes have one of the highest rates of gross primary production (*GPP*) and net primary production (*NPP*) of all ecosystems.

Carbon use efficiency (*CUE*) is the ratio of *NPP*:*GPP*. Freshwater marshes have a high *CUE*.

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Use your knowledge of *NPP* to explain why freshwater marshes have a high *CUE* and the advantage of this.

Do **not** refer to abiotic factors in your answer.

[2 marks]

Explanation _____

Advantage _____

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Freshwater marsh soils are normally waterlogged. This creates anaerobic conditions.

Use your knowledge of the nitrogen cycle to suggest why these soils contain relatively high concentrations of ammonium compounds and low concentrations of nitrite ions and nitrate ions.

[2 marks]

A student investigated the growth rate of a freshwater marsh plant.

The growth rate (R) of a plant can be determined using this equation.

$$R = \frac{(\ln W_2 - \ln W_1)}{t}$$

Where

$\ln$ = natural logarithm

t = duration of the investigation in days

W_1 = plant biomass at the start of the investigation

W_2 = plant biomass at the end of the investigation

The student used the equation above; however, she substituted height for biomass. This was because she did not want to destroy the plants to measure their biomass.

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State the assumption the student has made **and** suggest why this assumption might **not** be valid.

[2 marks]

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At the end of the investigation, the student noted the freshwater marsh plant had grown 268 mm in height, and now measured 387 mm. She calculated the rate of growth (R) to be $0.097 \text{ mm m}^{-1} \text{ day}^{-1}$

Use this information and, **substituting height for biomass**, use the equation to calculate the duration of the student's investigation.

Give your answer to the nearest full day. Show your working.

[2 marks]

_____ days

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